

P10:ECONOMIC FEASIBILITY OF TRANSITIONING FROM MANUAL TO AUTOMATED DISPENSING SYSTEMS (ADS) IN A MALAYSIAN STATE HOSPITAL: A 10-YEAR COST-SIMULATION ANALYSIS

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BACKGROUND

The public healthcare system in Malaysia relies heavily on decentralized outpatient pharmacy services to manage high patient volumes. Model is accessible but often may not cost-effectiveness due to high personnel. ADS offers a potential to improve productivity and reduce medication errors.^{1,2}

OBJECTIVE

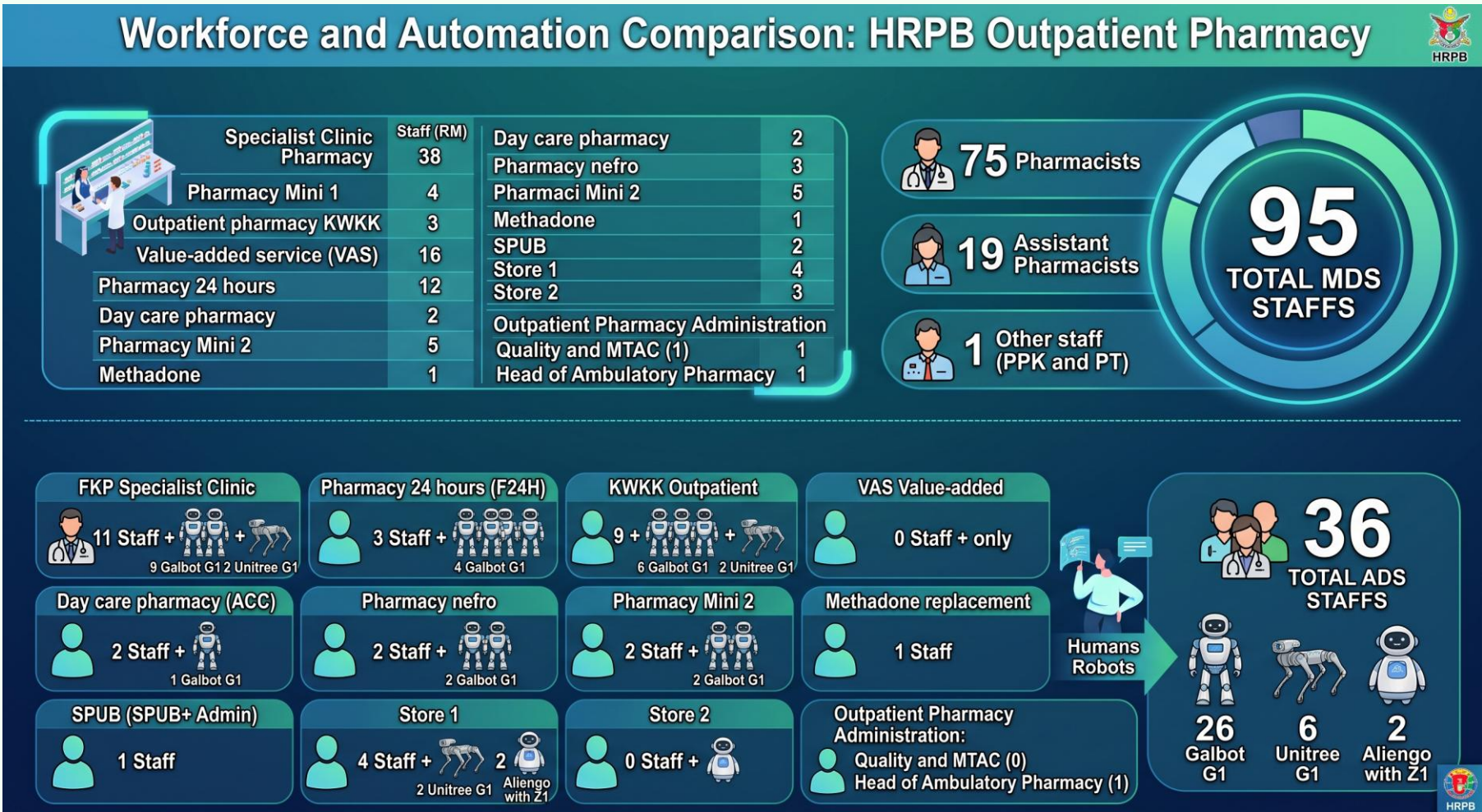
Explore economic feasibility of transitioning from Manual Dispensing System (MDS) to ADS model at Hospital Raja Permaisuri Bainun, comparing the cost per prescription over 10 years.

METHODOLOGY

Comparative cost simulation from healthcare payer perspective, bottom-up activity-based costing.

- ❖ Baseline (MDS): using retrospective January–September 2023 staff, prescription, and inventory data, incorporating standard Malaysian public salary and logistics scales.
- ❖ ADS Simulation: Using Galbot G1, Unitree G1 robotic Units and AI analytics especially at the two main outpatient pharmacy unit
- Capital expenditure: robotics with 5-year hardware replacement, renovation of the building
- System integration & software cost

RESULTS



COST PER PRESCRIPTION

MDS
RM 11.09/ prescription

ADS
RM 11.59/ prescription

TOTAL 10-YEAR COSTS

MDS :
RM78.75 million

ADS :
RM82.28 million

Broadly comparable although ADS requires large upfront investment:

- RM40.31 million robotics
- RM5 million renovation and software integration

SENSITIVITY ANALYSIS

- ❖ Highly sensitive to robot hardware prices
➔ +50% robotic cost ➔ RM12.98 /prescription
- ❖ Robust to modest staffing changes
➔ Increasing ADS workforce personnel from 36 to 45, the operational costs rise but remain less than 3 times from the base line

CONCLUSION

ADS model is economically feasible and offer superior operational value. Project to automate repetitive technical tasks may be reducing manpower & facilitating pharmacist redeployment to critical front-line clinical care & patient counselling.

REFERENCES

1. Are Decentralized Pharmacy Services the Preferred Model of Pharmacy Service Delivery within a Hospital? Can J Hosp Pharm. 2015;68(2):168-71.
2. Noparatayaporn P, Thaweethamcharoen T, Sakulbumrungsil R, Sangseenil W. Cost Comparison between the Manual System and Automated Dispensing Systems; How to Use Effective Health Technology at a Large Academic Hospital in Thailand. Value in Health. 2016;19(3):A31.